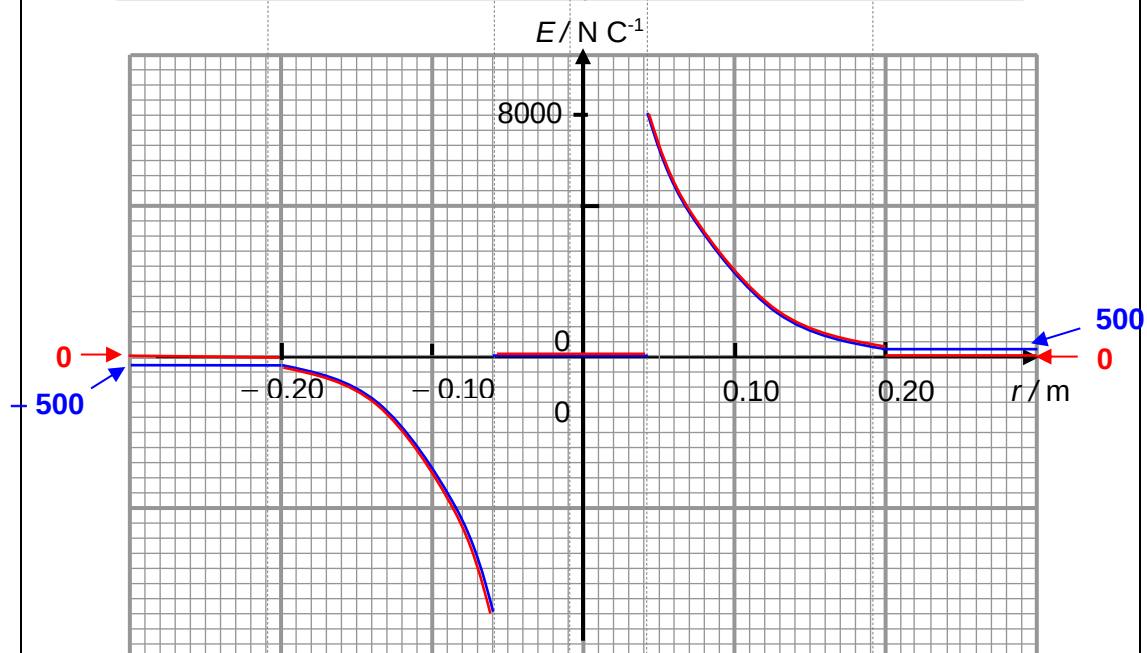
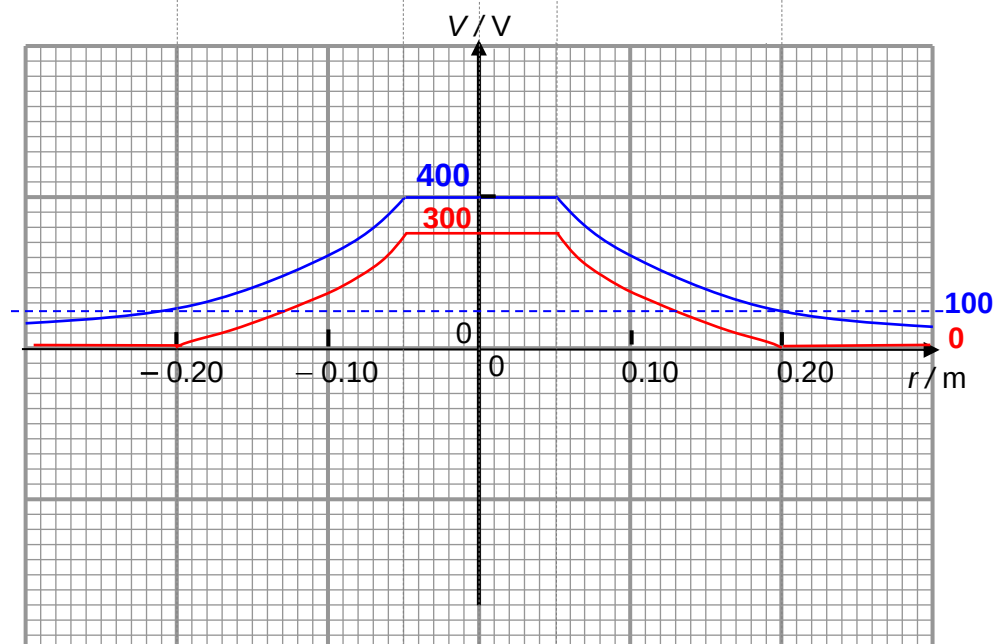
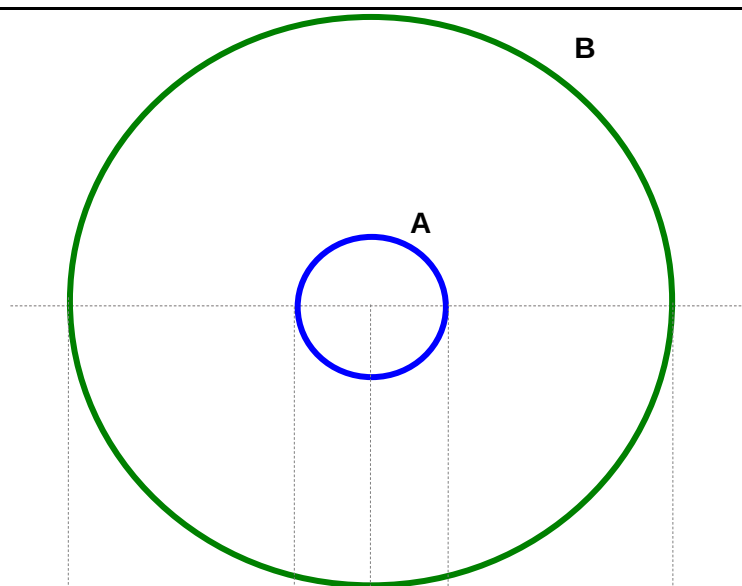


Q2	
(a)	Electric potential at a point in an electric field is defined as the <u>work done per unit positive charge by an external agent</u> in bringing a small test charge <u>from infinity to that point</u> , <u>without producing any acceleration</u> .
(b)	$V \propto \frac{1}{r}$ $\frac{V_2}{400} = \frac{0.050}{0.200}$ $V_2 = 100 \text{ V}$ $E \propto \frac{1}{r^2}$ $\frac{E_2}{8000} = \left(\frac{0.050}{0.200} \right)^2$ $E_2 = 500 \text{ N C}^{-1}$
(c)(i)1	Refer to sketches – $-R < r < R$: Horizontal line $V_{\text{inside}} = + 400 \text{ V}$ • $(r < -R)$ and $(r > R)$: Curve $V = +$ ○ $V_{0.050\text{m}} = + 400 \text{ V}$ $V_{0.200\text{m}} = + 100 \text{ V}$

(c)(i)2	$0 < r < R$: Horizontal line $E_{\text{inside}} = 0$ $(-R < r < 0)$ and $(R > r > 0)$: Curve & direction (When $r = +$, $E = +$) (When $r = -$, $E = -$) $E_{0.050\text{m}} = \pm 8000 \text{ N C}^{-1}$ $E_{0.200\text{m}} = \pm 500 \text{ N C}^{-1}$
(c)(ii)1	$-R < r < R$: $V_{\text{inside}} = + 300 \text{ V}$ (horizontal line) $(r < -R)$ and $(r > R)$: Curve $V = +$ $(r < -R)$ and $(r > R)$: horizontal line at $V_{0.200\text{m}} = 0$
(c)(ii)2	- Between $E_{0.050\text{m}}$ and $E_{0.200\text{m}}$: Curve & direction $(r = +, E = +)$ $(r = -, E = -)$ $E_{0.050\text{m}} = \pm 8000 \text{ N C}^{-1}$ $E_{0.200\text{m}} = \pm 500 \text{ N C}^{-1}$ $(r = +, E = +)$ $(r = -, E = -)$ $E_{\text{inside}} = 0$ $E_{\text{beyond } 0.200\text{m}} = 0$



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